

CRM PHASE 1

Test Coverage & Architecture Verification Report

Lifecycle / Pipeline Foundation

Review date: 17 September 2026

Basis: Binding CRM Target Model + completed Phase 1 execution plan

Review status: Architecture/test-gap assessment - source-code verification still required for flagged items

1. Executive Summary

Phase 1 is materially complete and the existing suite is green, but the available execution-plan evidence does not yet prove that every enterprise-grade invariant in the binding specification is covered. The main weakness is not test count; it is missing adversarial integration tests around the new pipeline persistence model, tenant isolation, tenant-safe references, and migration upgrades.

Area	Current evidence	Assessment	Priority
Lifecycle rename	Existing tests updated; fresh DB migration path green	Mostly covered; legacy-row upgrade test missing	P1
Pipeline domain rules	4 in-memory/domain tests	Useful but insufficient for persistence/DB invariants	P0
Pipeline RLS	No explicit table-by-table pipeline RLS test evidenced	Gap	P0
Tenant-safe FK/integrity	Schema configured with composite FKs	Negative cross-tenant tests not evidenced	P0
Opportunity version-stage consistency	No explicit invariant test evidenced	Potential design defect until proven	P0
Party migration	Fresh tip migration + one MasterData readability test	Real legacy backfill upgrade test missing	P1
Target stage capability model	Implementation shows Name + SortOrder only	Binding-model gap or undocumented deferral	P0/P1

Important limitation: this report is based on the binding specification and the completed Phase 1 execution-plan record. It does not claim that a test is absent if it exists in the repository but is not represented in those materials. Each P0 item should therefore be verified against the actual test/source tree before remediation.

2. Review Scope and Sources

- Enterprise CRM Target Model - Binding Implementation Specification
- CRM Phase 1 - Lifecycle/Pipeline Foundation - Execution Plan
- Recorded post-merge verification: clean build, 46 CRM tests + 30 MasterData tests green
- Review focus: lifecycle invariants, pipeline rules, tenant boundaries, persistence, migration behavior, target-model alignment

3. What the Existing Tests Prove

- The canonical lifecycle naming is now Draft / Open / Won / Lost.
- Existing lifecycle behavior survived the mechanical rename without regressions.
- Pipeline definition/version/stage domain objects reject duplicate version numbers and duplicate sort order in the same version.
- Different versions can reuse stage names/sort orders, supporting version independence.
- CRM and MasterData test projects run together after the PartyRef cutover.
- Opportunity tenant isolation tests remain present after the old CRM Party entity was removed.

However, a green suite does not prove an invariant that no test actively attempts to violate.

4. Detailed Findings

F-01 [P0] Pipeline hierarchy persistence round-trip is not evidenced

Observation: The plan documents in-memory domain tests for PipelineDefinitionVersion, but no test that creates a PipelineDefinition, adds a version and stages through the public domain API, persists the graph to PostgreSQL, clears the EF tracker, and reloads the data.

Why it matters: This is especially important because Versions/Stages navigation properties were deliberately ignored by EF after a duplicate/shadow-FK defect was found. The documented domain constructors also do not visibly assign parent IDs in the shown code.

Expected verification: Create -> persist -> clear tracker -> reload -> assert Definition/Version/Stage IDs, tenant IDs and relationships. If this fails, classify as implementation/design defect, not merely a missing test.

F-02 [P0] Pipeline tables need explicit non-superuser RLS integration tests

Observation: The binding specification requires all new persistence to preserve RLS under a non-superuser runtime role. The available plan shows the existing TenantIsolationTests were retargeted to Opportunity, not explicit tests for all three new pipeline tables.

Why it matters: Application-level filtering is not an enterprise tenant boundary. A missed WHERE clause must still be blocked by PostgreSQL.

Expected verification: For pipeline_definitions, pipeline_definition_versions and pipeline_stages: Tenant A cannot SELECT, UPDATE or DELETE Tenant B rows; attempts to create rows under another tenant are rejected.

F-03 [P0] Cross-tenant referential integrity needs negative tests

Observation: The EF mappings are described as tenant-safe composite foreign keys, but the plan does not show tests that intentionally cross tenant boundaries.

Why it matters: Configuration can regress silently, as already happened once when duplicate/shadow FKs were generated.

Expected verification: Tenant A version cannot reference Tenant B definition. Tenant A stage cannot reference Tenant B version. Tests must fail at the DB boundary.

F-04 [P0] Opportunity PipelineDefinitionVersion / PipelineStage consistency is not proven

Observation: Opportunity stores both PipelineDefinitionVersionId and PipelineStageId. The available plan does not show a test that prevents a Stage from Version B being paired with Version A.

Why it matters: If both foreign keys are independently valid but not correlated, the row can be internally inconsistent while satisfying both individual FKs.

Expected verification: A DB/domain invariant must guarantee that the selected stage belongs to the selected version. Add a negative integration test first; if it passes incorrectly, fix the model.

F-05 [P1] Database uniqueness constraints need integration-level negative tests

Observation: The model declares unique indexes for tenant pipeline name, definition-version number, stage sort order and stage name, but only some invariants are tested in memory.

Why it matters: Domain guards and DB constraints protect different entry paths. Imports, migrations and future integrations can bypass aggregate methods.

Expected verification: Test same-tenant duplicates rejected; same values across different tenants/versions allowed where intended.

F-06 [P1] Lifecycle migration is not tested as an upgrade with pre-existing legacy rows

Observation: Fresh Testcontainers migration runs prove a clean database can reach the latest schema. They do not prove existing waiting/offered/completed/canceled rows are transformed correctly.

Why it matters: The migration contains explicit data remapping logic. That logic needs a database-at-N-1 upgrade test.

Expected verification: Migrate to pre-rename migration, seed all four legacy statuses, migrate to latest, assert draft/open/won/lost and renamed date/reason fields.

F-07 [P1] CRM Party -> MasterData backfill is not validated with real legacy rows

Observation: The planned row-count/backfill verification was reduced to one test after crm.parties disappeared at the migration tip. The remaining test proves MasterData is readable, not that legacy CRM rows are copied correctly.

Why it matters: A destructive cutover should be verified from the previous migration state, especially for row identity and tenant preservation.

Expected verification: Start at the legacy migration, seed crm.parties, execute backfill/drop sequence, then verify id, tenant_id, name, surname, phone, email and party_type mapping in masterdata.parties.

F-08 [P0/P1] Implemented PipelineStage shape is narrower than the binding target model

Observation: The binding model expects tenant-defined stage name/order plus active state and allowed capabilities/policy references, with stable IDs/keys. The shown implementation contains Id, TenantId, PipelineDefinitionVersionId, Name, SortOrder and CreatedAt.

Why it matters: The target model explicitly says behavior must not depend on stage-name strings and gives capability examples such as quote.create, quote.send, discount.request and approval.request.

Expected verification: Verify whether stable key/code, active, and capability/policy references were intentionally deferred by a recorded architecture decision. If not, treat this as a Phase 1 foundation gap before Phase 2 uses stages for behavior.

F-09 [P1] Lifecycle test suite should be mapped against a full transition matrix

Observation: The plan states existing lifecycle tests were mechanically renamed, but no explicit matrix proves every legal and illegal transition is covered.

Why it matters: Renamed tests can preserve old assumptions while missing new target semantics such as terminal Won/Lost and policy-controlled Reopen.

Expected verification: Produce a state x command matrix and link each allowed/denied transition to an exact test. Missing cells become new tests or explicitly deferred commands.

F-10 [P2] Completion/CI evidence in the plan should be normalized

Observation: The plan header records Phase 1 as merged to main with post-merge verification, while older acceptance notes still contain branch-era wording about CI not having run.

Why it matters: Stale evidence creates ambiguity in release readiness and weakens the document as an audit trail.

Expected verification: Update the final acceptance block to reflect the true main-branch/CI state and remove superseded branch-status statements.

5. Priority and Remediation Order

Recommended order before Phase 2:

1. P0-1: Add the pipeline persistence round-trip test before changing any implementation.
2. P0-2: Add RLS isolation tests for all three pipeline tables using the non-superuser runtime role.
3. P0-3: Add cross-tenant FK negative tests.
4. P0-4: Add Opportunity Version/Stage mismatch test; fix the data model if it currently allows mismatch.
5. P0-5: Reconcile PipelineStage target shape (stable key, active, capability/policy refs) with the binding decision record.
6. P1-1: Add real migration upgrade tests for lifecycle remap and Party backfill.
7. P1-2: Add DB uniqueness tests and complete the lifecycle transition matrix.
8. P2: Normalize stale CI/completion documentation.

6. Proposed Adversarial Test Pack

Test	Purpose	Priority
T-01 Pipeline graph persists/reloads	Create definition/version/3 stages via domain API; SaveChanges; clear tracker; reload; assert full relationship integrity.	P0
T-02 Definition RLS cross-tenant block	Tenant A runtime role cannot read/update/delete Tenant B definition.	P0
T-03 Version RLS cross-tenant block	Same for version table.	P0
T-04 Stage RLS cross-tenant block	Same for stage table.	P0
T-05 Version -> foreign-tenant definition rejected	Composite FK must reject cross-tenant parent.	P0
T-06 Stage -> foreign-tenant version rejected	Composite FK must reject cross-tenant parent.	P0
T-07 Opportunity stage/version mismatch rejected	Selected stage must belong to selected version.	P0
T-08 Duplicate pipeline name / same tenant rejected	DB uniqueness.	P1
T-09 Same pipeline name / different tenant allowed	Tenant-scoped uniqueness.	P1
T-10 Duplicate version number rejected at DB	DB uniqueness independent of domain guard.	P1
T-11 Duplicate stage name rejected at DB	DB uniqueness.	P1
T-12 Duplicate sort order rejected at DB	DB uniqueness.	P1
T-13 Legacy lifecycle upgrade maps all statuses	N-1 database with four legacy states migrates correctly.	P1
T-14 Legacy Party backfill preserves identity/tenant	Real crm.parties rows survive cutover into MasterData.	P1

7. Binding Acceptance Criteria vs. Evidence

Binding expectation	Current evidence	Gap / next proof
Tenant can operate with 1..N Open stages without code changes	Partially evidenced by additive model/domain tests	Needs persistence/API behavior proof in Phase 2
Two tenants can use different stage names/order/capabilities	Names/order structurally possible	Capabilities not present in shown Stage model; requires decision
Changing pipeline creates new version and does not silently remap open opportunities	Version objects and independent versions tested	No persisted-open-opportunity remap test yet
No stage-name hardcoding for behavior	Architectural rule documented	Requires code scan + capability-key design before behavior is added
All new persistence enforces RLS under runtime role	Existing CRM RLS discipline present	Pipeline table-specific tests not evidenced
Tenant-safe referential integrity	Composite FK configuration documented	Cross-tenant negative tests not evidenced
State-changing commands authorized/concurrency-safe/idempotent/evidence/outbox	Out of Phase 1 by design	Must be proven in Phase 2

8. Release Gate Recommendation

Do not treat Phase 1 as fully verified for an enterprise foundation until the P0 items are either:

- Proven by existing source/test evidence, or
- Covered by new adversarial tests, with any exposed design defect corrected.

This does not mean the Phase 1 implementation is fundamentally wrong. It means the current evidence is stronger on happy-path/domain behavior than on persistence and tenant-boundary failure modes. Given the project's foundation-first standard, those failure modes should be made explicit before Phase 2 begins using pipeline stages as business-control inputs.

9. Suggested Claude Review Prompt

Phase 1'i tekrar implement etme.

Mevcut Phase 1 implementation ve test suite üzerinde adversarial test-gap review yap.

Özellikle:

1. PipelineDefinition -> PipelineDefinitionVersion -> PipelineStage persistence round-trip.
2. Her uç pipeline tablosu için non-superuser RLS SELECT/UPDATE/DELETE/INSERT boundary tests.
3. Cross-tenant composite FK negatif testleri.
4. Opportunity PipelineDefinitionVersionId / PipelineStageId tutarlılığı.
5. DB unique constraint negatif testleri.
6. N-1 -> latest lifecycle migration upgrade testi (waiting/offered/completed/canceled remap).
7. N-1 -> latest crm.parties -> masterdata.parties gerçek backfill testi.
8. Binding spec Stage modelindeki stable key/code, active, capability/policy refs ile mevcut implementation'ı karşılaştır.
9. Draft/Open/Won/Lost x command transition matrix çıkar ve her hücreyi exact test ile eşleştir.

Önce raporla; implementation yapma.

'All tests green' ifadesini kalite kanıtı olarak kabul etme.

Her invariant için onu gerçekten kırmaya çalışan negatif test olup olmadığını göster.

10. Conclusion

Overall assessment: Phase 1 is structurally advanced and has a healthy regression suite, but the evidence should be strengthened before it becomes the permanent foundation for Phase 2. The highest-risk gaps are pipeline persistence round-trip, pipeline RLS, cross-tenant referential integrity, and Opportunity stage/version consistency. These are precisely the kinds of defects that can remain invisible while all happy-path tests stay green.

Appendix A - Source Trace

- Binding Specification §4: canonical lifecycle; Won/Lost terminal by default; Reopen requires explicit command/reason/policy/evidence.
- Binding Specification §5: pipeline is tenant-configurable and versioned; stage behavior uses stable IDs/capability-policy keys; allowed capabilities include quote.create/quote.send/discount.request/approval.request.
- Binding Specification §22-24: preserve RLS, tenant-safe FKs, concurrency, idempotency, evidence and outbox; add tests for lifecycle invariants, pipeline rules and tenant boundaries.
- Phase 1 Execution Plan Task 2: four pipeline domain tests, composite FK mappings, Versions/Stages EF Ignore() decision.
- Phase 1 Execution Plan Task 3: PartyRef cutover, reduced backfill verification, retargeted TenantIsolationTests, recorded migration-order deviations.
- Phase 1 Execution Plan Acceptance: 46 CRM + 30 MasterData tests green after post-merge verification.